INFORMATION SECURITY

Seungchan Song



Attack it first.
Prove the risk in numbers.
Work out what to fix first.

I learned how attackers think through system hacking, then built RAG-DIAG, a tool that reproduces and measures where and how much personal data leaks from AI (RAG) services. I want to plan privacy protection and security services and strategy on evidence and numbers, not on intuition.

Education	Sejong Univ., Information Security Expected Feb 2027 · GPA 4.10 / 4.5 · 130 credits	Email	chanchan3314@naver.com
Credentials	Engineer Information Security · TOEIC 850 · OPIc IH	Phone	010-7406-3317
Programming	Python main · C	Born	2001.01.04
Languages	English proficient · 中文 intermediate	GitHub	github.com/seungchan-song

3rd Encouragement Award
Capstone competition (RAG-DIAG)

Patent 1 filed
Filed 2026.08.31 · pending examination

Elite 3 Distinction
P4C system-hacking track · top 3 on the completion exam

ABOUT

Three things about how I work

01

I measure risk

I replace qualitative claims with metrics. In RAG-DIAG I designed the risk-score formula, the control-group incremental measurement and the four-stage Korean PII detection pipeline.

02

I learn in depth

I entered Sejong University's Information Security department with no coding background, then became the department's top student, finished a system-hacking track as Elite 3 and earned the Engineer Information Security certification, one after another.

03

I move teams

I led the capstone team and a hackathon team at UBC. Five and a half years at an international school in China made working in English natural; my Chinese is intermediate, which I used to talk with a Chinese teammate at UBC.

FEATURED PROJECT

RAG-DIAG

A CLI that diagnoses personal-data leakage in RAG systems

Team lead · team of 4 | Sejong Univ. capstone | 2026.03 – 2026.09 | MIT License

GitHub ↗ | PII NER model ↗

MY ROLE

As team lead I ran a four-person team and designed how the diagnosis judges risk.

01
Team lead
Ran the meetings, gave each teammate work that fit their role, and gave feedback every time.

02
Korean PII detection pipeline
Designed the four-stage structure and the judging rules.

03
Risk formula and control-group experiment
Built a way to measure only the leakage the attack adds.

04
Documentation lead
Led the writing of four documents: requirements analysis, project design, final report and patent application.

APPROACH

In one line · throw attack questions at a RAG system, then score how much personal data leaks in its answers.

STEP 1

Ask questions

- R2 Retrieved-data leak
- R4 Membership inference
- R7 System-prompt leak
- R9 Indirect prompt injection

Control group ordinary questions with no attack, asked alongside as the baseline.

→

STEP 2

The target RAG answers

A RAG built over indexed documents answers both the attack and the control questions; model, Top-k and reranker can be changed per run.

Haystack v2 | FAISS | BGE-m3-ko | Qwen2.5-3B · Ollama

→

STEP 3

Find the leaked PII and score it

- Detect — Regex › checksum › NER › sLLM
- Score — 50% success rate + 50% severity
- Compare — Increase over the control
- Output — HTML · CSV · JSON

RESULT

Under what conditions were these results obtained?

Target	**The RAG built into RAG-DIAG** · Haystack v2 · FAISS · Qwen2.5-3B run locally with Ollama
Data	**Synthetic documents** seeded with **fake** personal data (ID numbers, cards, accounts, contacts). No real personal data.
Runs	**4 attacks + a control group** · 988 queries · reranker on/off · run on 2026-08-14, 0 failures

What I found

Q1 Which attack was riskiest?

The attack that succeeds most often is not the most damaging.

Each attack was scored out of 100 from its success rate and how deep it went. System-prompt leakage had the lowest success rate (15.0%) but exposed 83% of the defensive rules when it worked, so it ranked first.

Data
Risk score (of 100): system-prompt leak 49 · membership inference 40 · indirect injection 39 · retrieved-data leak 37

Q2 Did the attack leak more than ordinary questions?

What an attack adds only shows up with a control group.

With question counts matched, the retrieved-data-leak attack exposed twice as much ID, card and account data as ordinary questions. Without a control group you cannot separate what leaks anyway from what the attack adds.

Data
Unique-ID and financial PII exposed: control 27 → attack 54

Q3 Does turning on a defence make it safe?

A defence that blocks one attack can make another easier.

Turning on a reranker, which re-filters retrieved passages, reduced retrieved-data leakage but increased membership inference. A defence has to be measured in your own environment before you adopt it.

Data
Reranker off → on, successful attacks: retrieved-data leak 37 → 12 · membership inference 15 → 46

Quality checks · all 443 regression tests passing · PII detector recall 94.5%, precision 79.7% (1,320-document synthetic hold-out set)

OUTCOME

Award	**Capstone competition, 3rd place (Encouragement Award)**
Patent	**1 patent application** · “System and method for diagnosing information-leakage vulnerabilities in retrieval-augmented generation” No. 10-2026-0164171 · filed 2026.08.31 · pending, not yet granted · applicant: Sejong Univ. Industry-Academic Cooperation Foundation
Contest	**Entered the 2026 Open Source Developer Contest (student division)**
Release	**Open-sourced** · code (MIT License) plus the models and dataset on GitHub and Hugging Face

OTHER WORK

Other work

SYSTEM HACKING

P4C system-hacking track

2025.06.09 – 2025.08.31

Top 3 on the completion exam (Elite 3). Survived a program with a 31.8% completion rate and earned the Elite 3 distinction given to the top three on the final exam. Reproduced memory exploits through to bypassing protections such as SROP. I started with no Linux background and nearly dropped out after failing early assignments; I changed how I studied (map how the concepts connect first, then reproduce them hands-on) and finished in the top three.

BLOCKCHAIN · HACKATHON

UBC Blockchain Summer Institute

2026.08.16 – 2026.08.30

Hackathon 3rd place, team lead. With Swiss and Chinese teammates, designed an on-chain permission list and tamper-proof access log for medical records. No patient data goes on the chain, only who was allowed in and who tried.

GitHub ↗

PYTHON · TEAM LEAD

Voice Phishing Radar

2026.09.21 – 2026.10.02

SK Shieldus Rookies team project, team lead of 6. Combines police statistics and news to show damage trends and new scam methods as a web dashboard, alert mail and Notion log. I built the shared code and the Flask web app and managed integration and the repository.

GitHub ↗

SKILLS

Languages & tools I have used

Only things I have actually used in projects, coursework and labs.

Programming languages

MAIN

Python

RAG-DIAG · Voice Phishing Radar

C

System-hacking practice · coursework

Foreign languages

PROFICIENT

English

TOEIC 850 · OPIc IH · 5.5 years at an international school in China · English teamwork at UBC

INTERMEDIATE

中文

Clearly behind my English. Lived in China · used it with a Chinese teammate at UBC

Tools & technologies

RAG · LLM

Haystack v2 · FAISS · Ollama · Qwen2.5-3B · BGE-m3-ko · Hugging Face · KPF-BERT (NER)

WEB · DATA · COLLABORATION

Flask · pandas · matplotlib · BeautifulSoup · MongoDB Atlas · Notion API · Typer · Rich (CLI) · Git · GitHub · Notion

SECURITY · SYSTEMS

Linux · Memory exploitation · SROP · Proxmox · Docker · QEMU · Forensics (FTK Imager · HxD)

BLOCKCHAIN

Ethereum (Sepolia) · MetaMask

PROFILE

Education, training & credentials

2017.03 – 2020.02
Education **Sangsan High School**

2021.03
Education **Entered Sejong University, Information Security**
Started with no coding background and studied Python and C side by side.

2023.04 – 2025.01
Service **Republic of Korea Air Force · anti-aircraft gun operation**
Model soldier: mediated between about 60 soldiers and officers (6 months)

2025.06.09 – 2025.08.31
Training **P4C cohort 13, system-hacking track**
RED course · 31.8% completion rate · top 3 on the completion exam · Elite 3 certificate of excellence

2026.02.28
Credential **TOEIC · 850**
LC 465 · RC 385

2026.08.16 – 2026.08.30
Training **UBC Blockchain Summer Institute**
University of British Columbia, Canada · blockchain technology and regulation · hackathon 3rd place (team lead)

2026.08.28
Credential **Engineer Information Security**
Korean national technical qualification

2026.09.27
Credential **OPIc · IH (Intermediate High)**
ACTFL English speaking assessment

2026.09 –
Training **SK Shieldus Rookies, cohort 35**
Cloud-based smart convergence security course · in progress

2027.02 expected
Education **Sejong University, Information Security: graduation**
Now in the final (4th-year, 2nd) semester. Returned as a junior after military service and finished the first semester back as the department's top student.

Credits **130** | GPA **4.10 / 4.5**

CONTACT

I'd like to explain privacy and security risk with evidence and numbers, together with your team.

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GitHub ↗ | PDF (Korean) | PDF (English summary)